Problem Set 3

Version of September 17, 2026

Problem 3.1 (Domain estimation). We use the data set `sepe.dat` from Problem Set 1. Treat the data as a simple random sample without replacement (SRS) of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} identified by `code == 1`.

a) Assume that the domain size N_{31} is unknown. Estimate the domain population mean $\bar{y}_{U31}$ of environmental-protection expenditures and its standard error.

> **Hint.** If `dn` is a survey design created with `svydesign()`, use `subset(dn, code == 1)` to restrict the design to the domain. Do not create a new SRS design from the domain observations: under the original SRS, the domain sample size n_{31} is random.

b) Now assume that the domain size $N_{31} = 650$ is known, but that this information was not used when the sample was drawn. Estimate the population domain total of environmental expenditures $t_{y,31}$ and its standard error.

Problem 3.2 (Planning a stratified sample). We *plan* a survey to estimate the population mean of environmental expenditures (`expend`). We want to assess whether stratification by industry group (`code`) can improve precision relative to a simple random sample without replacement (SRS).

We use the data set `sepe.dat` from Problem Set 1. Treat the data as an SRS of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} of size $N_{31} = 650$, identified with `code == 1`. Assume that:

- the planned net sample size is $n = 800$ establishments;
- the anticipated response rate is 80 % in both strata; and
- the within-stratum variances of `empl` and `expend` can be estimated from the existing SRS `sepe.dat`.

a) Allocate the net sample *proportionally* to the two strata. Determine the required gross sample sizes and estimate the variance of the stratified estimator of the population mean of expenditures.

b) Now use a Neyman-type (optimum) allocation, but base the allocation on the within-stratum standard deviations of the auxiliary variable `empl` rather than on `expend`.
   - Determine the required gross sample sizes.
   - Under what conditions is it reasonable to derive the allocation from an auxiliary variable?
   - Estimate the variance of the stratified estimator of mean environmental expenditures under this allocation.

c) Compare the estimated variances from parts a) and b) with the estimated variance under a simple random sample of $n = 800$. Does stratification improve precision?

Problem 3.3 (PPS Sampling and the Horvitz–Thompson estimator). We use the data set `sepe.dat` from Problem Set 1 and treat the $N = 800$ establishments as the finite population U. We want to draw a PPS sample without replacement of size $n = 200$, using the number of full-time employees as the measure of size. The data set contains:

`empl`: number of full-time employees, denoted by x_i. These values are known from a register before sampling;

`expend`: environmental expenditures, denoted by y_i. We treat these values as unknown until an establishment is sampled.

a) Compute the first-order inclusion probabilities and draw a PPS sample of $n = 200$ establishments without replacement using Brewer's method. If there were no certainty units, the inclusion probabilities would be

$$\pi_i = \frac{n x_i}{t_x}, \qquad t_x = \sum_{i \in U} x_i.$$

Compare these naive probabilities with the probabilities returned by `pps_probabilities()` of the `robsurvey` package. What do you observe?

Hint. Attach the packages `survey` and `robsurvey` to the search path. Compute the inclusion probabilities and draw the sample as follows:

```
pps_setup <- pps_probabilities(size = sepe$empl, n = 200)
pi <- as.numeric(pps_setup)

set.seed(11)
s <- pps_draw(pps_setup, method = "brewer")
```

b) Create a survey design object for the sample from part a). Then use `svyplot()` to plot environmental expenditures against the number of employees. The bubble sizes represent the sampling weights $w_i = 1/\pi_i$. Describe the main features of the plot.

Remarks. For the PPS design of part a), use

```
dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                     data = sepe[s, ])
```

Here, `fpc = ~pi` supplies the first-order inclusion probabilities. The option `pps = "brewer"` tells `survey` to use Brewer's approximation for the second-order inclusion probabilities needed for variance estimation. This is distinct from Brewer's sampling algorithm used in part a).

c) Estimate the population total of environmental expenditures,

$$t_y = \sum_{i \in U} y_i,$$

using the Horvitz–Thompson estimator and the design from part b). Also compute an estimate of the standard error of the estimator.